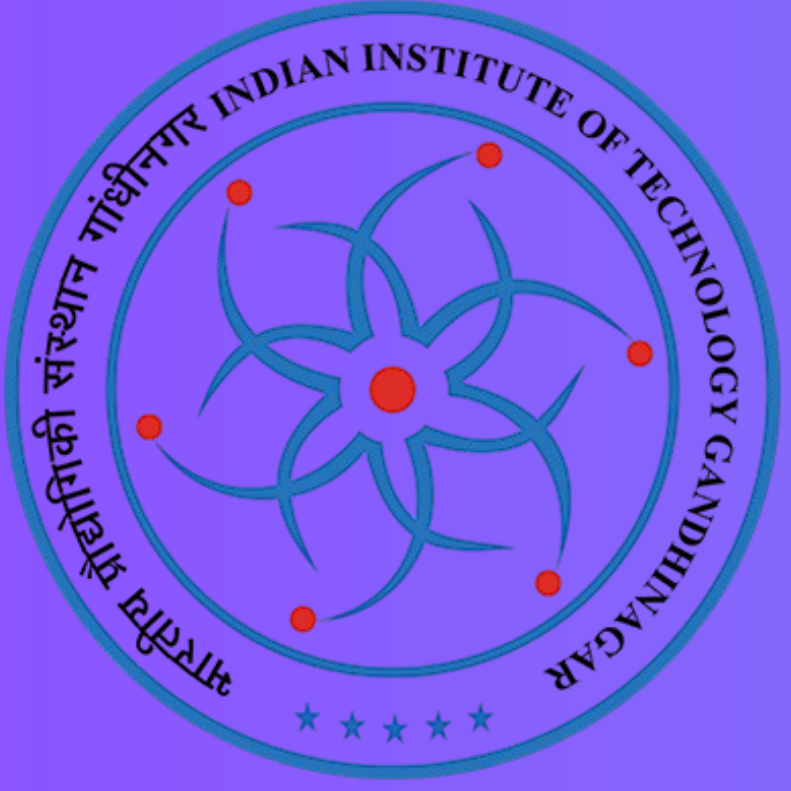


Accelerating Post Quantum Cryptography Algorithms on FPGA



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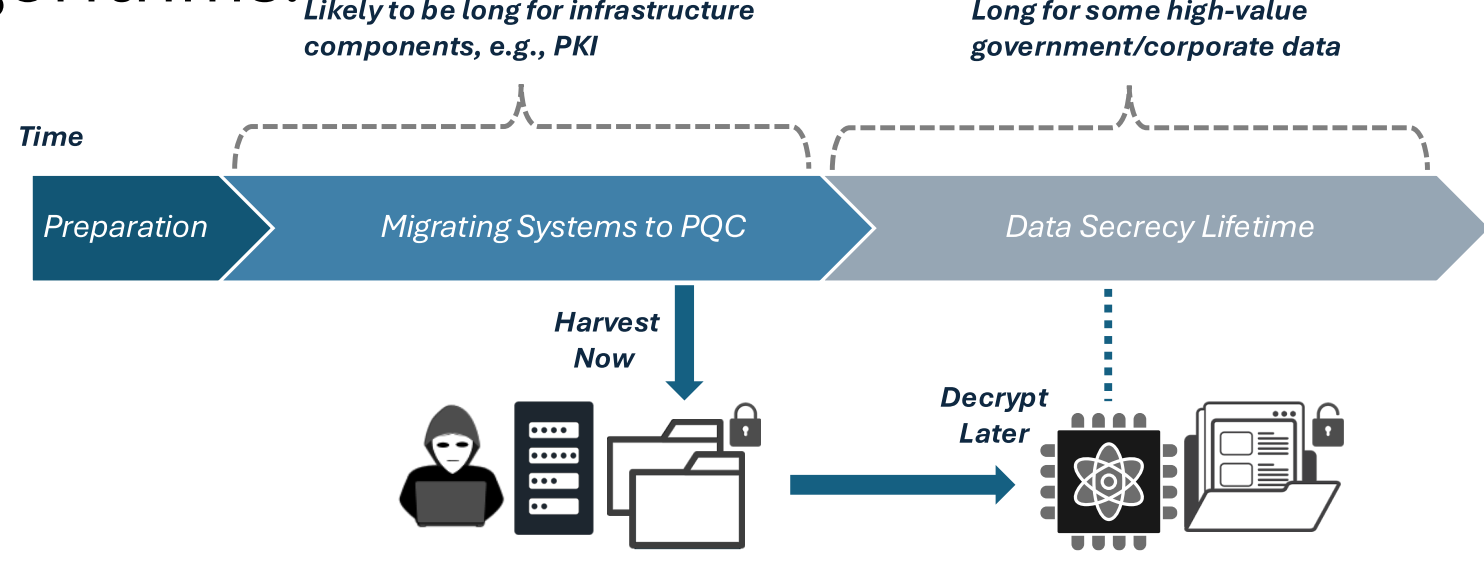
Advisor: Prof. Joycee Meckie

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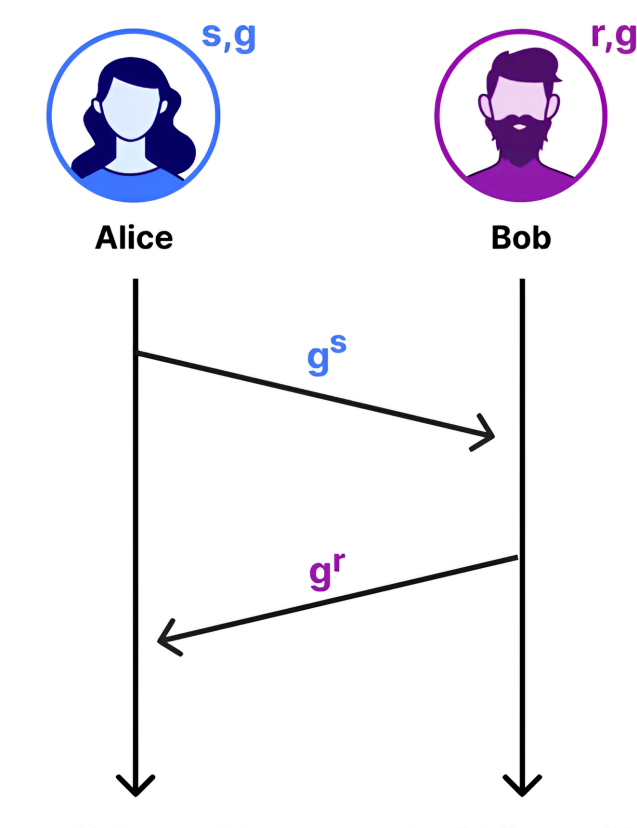
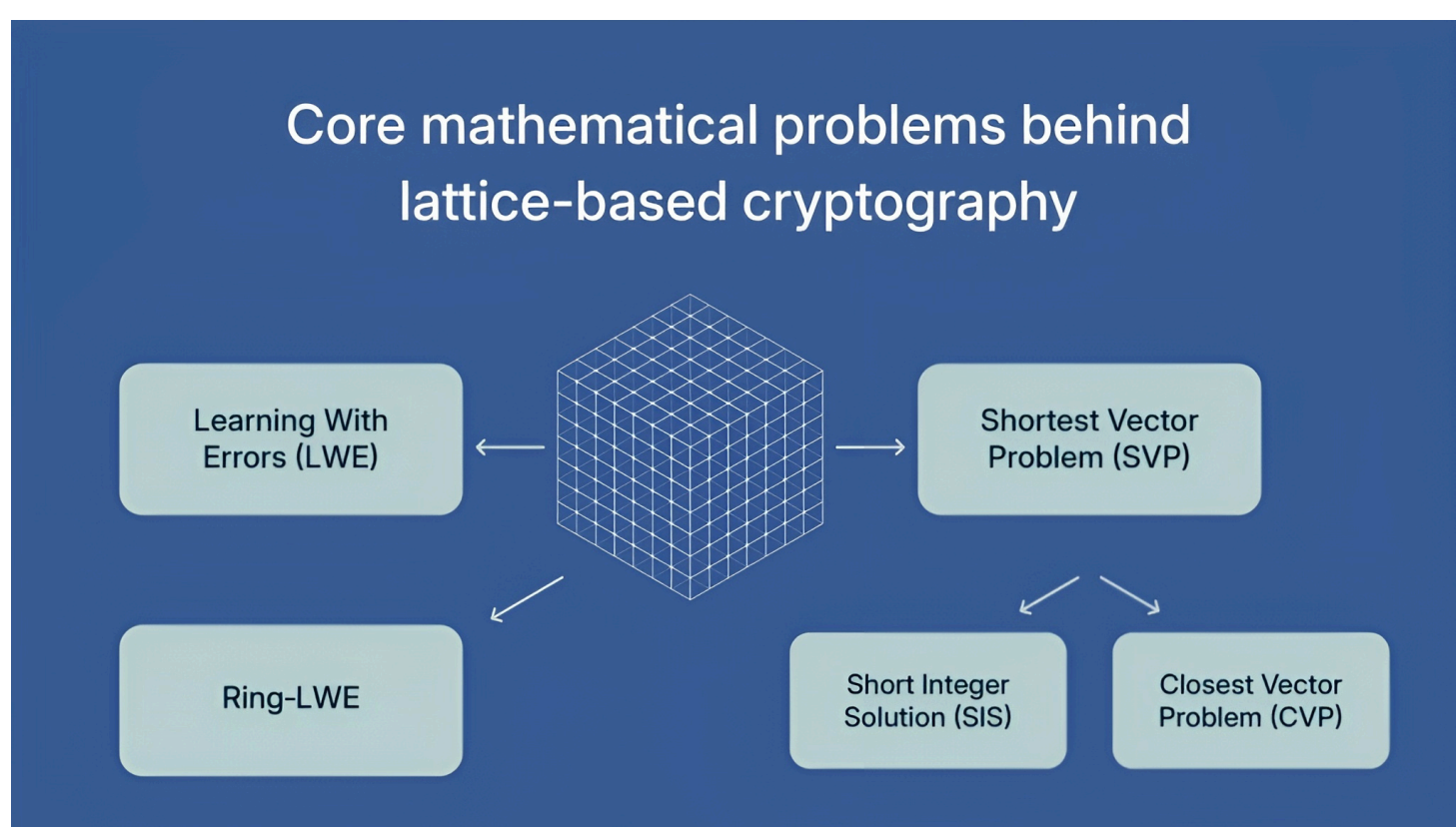


Motivation and Problem Statement

- Quantum computing can break present crypto algorithms.



- Most PQC algorithms use lattice based cryptography.
- Bottleneck:** Polynomial Multiplication is computationally expensive.
- Software Limit:** NTT reduces the computation, but software is too slow for the edge.
- Objective:** Hardware acceleration on FPGA for NTT.

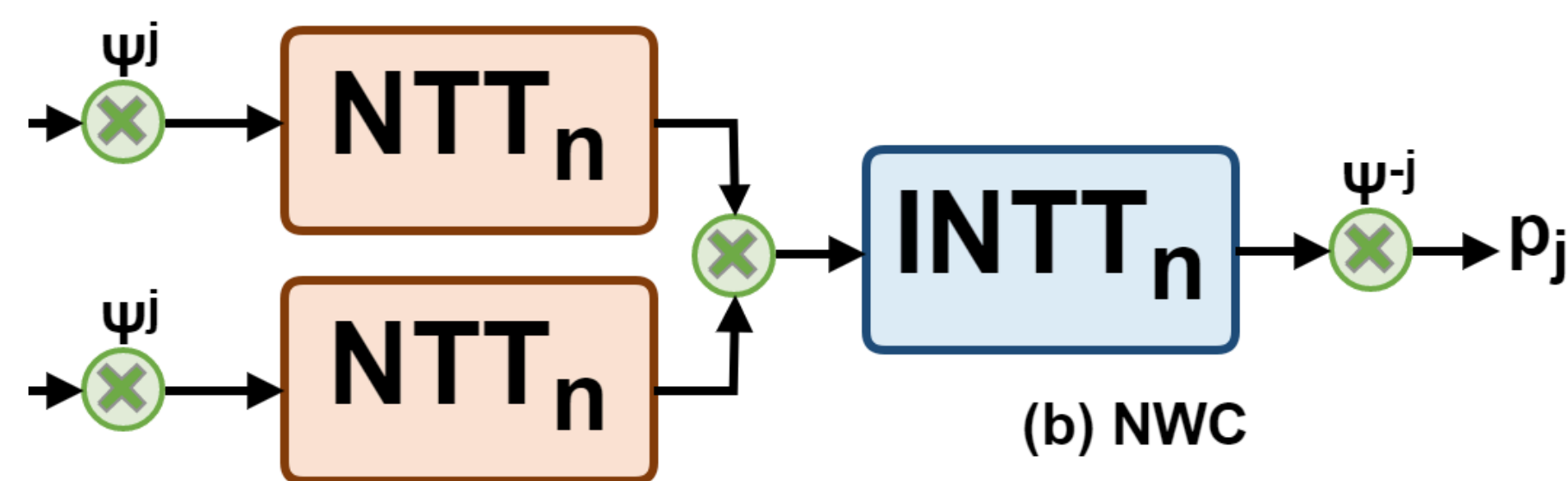


Algorithm for NTT

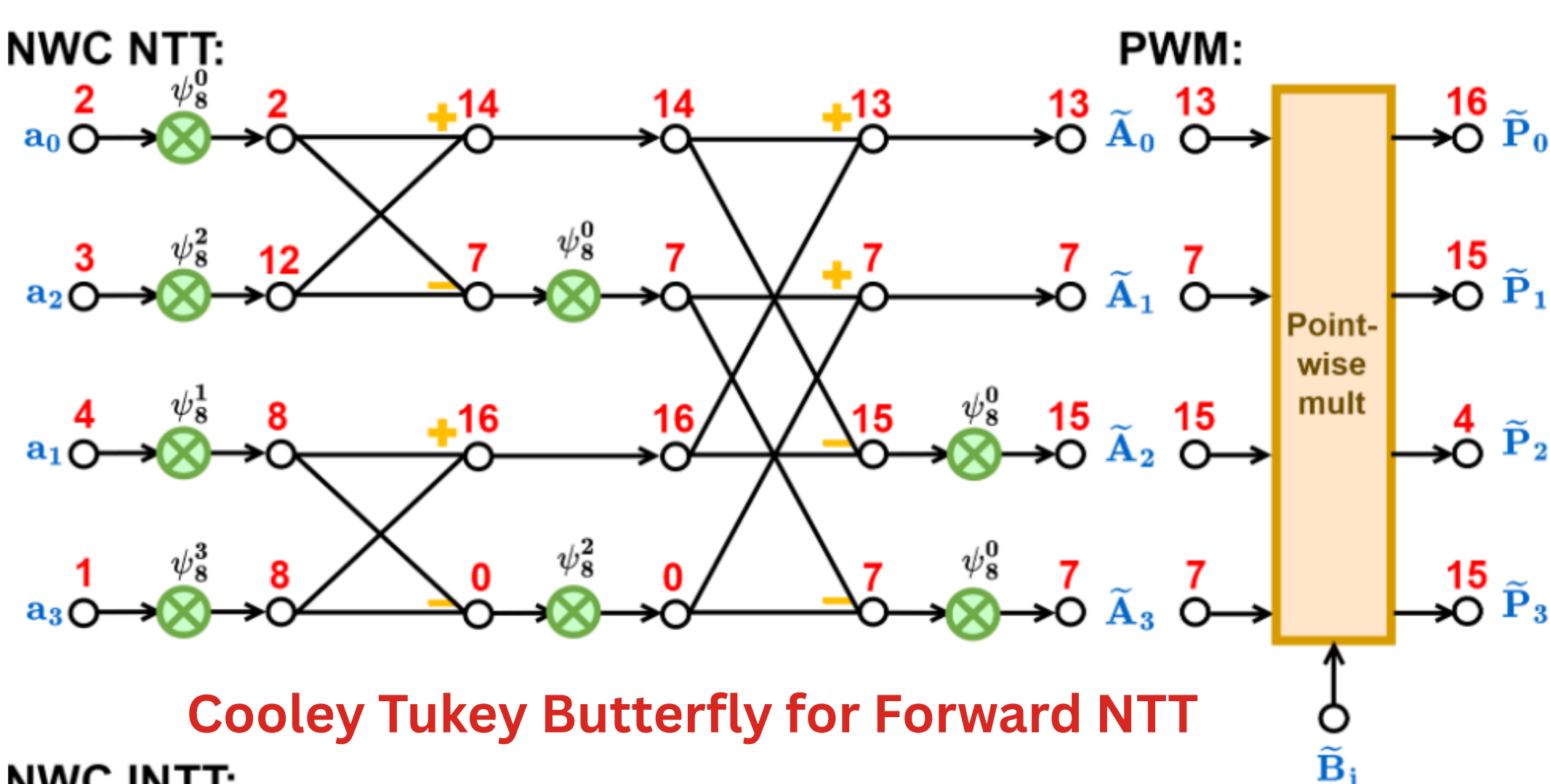
Example showing modulo reduction

- Standard Multiplication**
 $(1 + 2x + 3x^2)(4 + 5x + 6x^2)$
 $= 4 + 13x + 28x^2 + 27x^3 + 18x^4$
- Polynomial Reduction**
Apply $x^3 \equiv 1$ (Wrap-around)
 $27x^3 \rightarrow 27$ (adds to constant)
 $18x^4 \rightarrow 18x$ (adds to x)
 $= (4 + 27) + (13 + 18)x + 28x^2$
 $= 31 + 31x + 28x^2$
- Coefficient Reduction**
Modulo $q = 7$
 $31 \equiv 3 \pmod{7}$
 $28 \equiv 0 \pmod{7}$

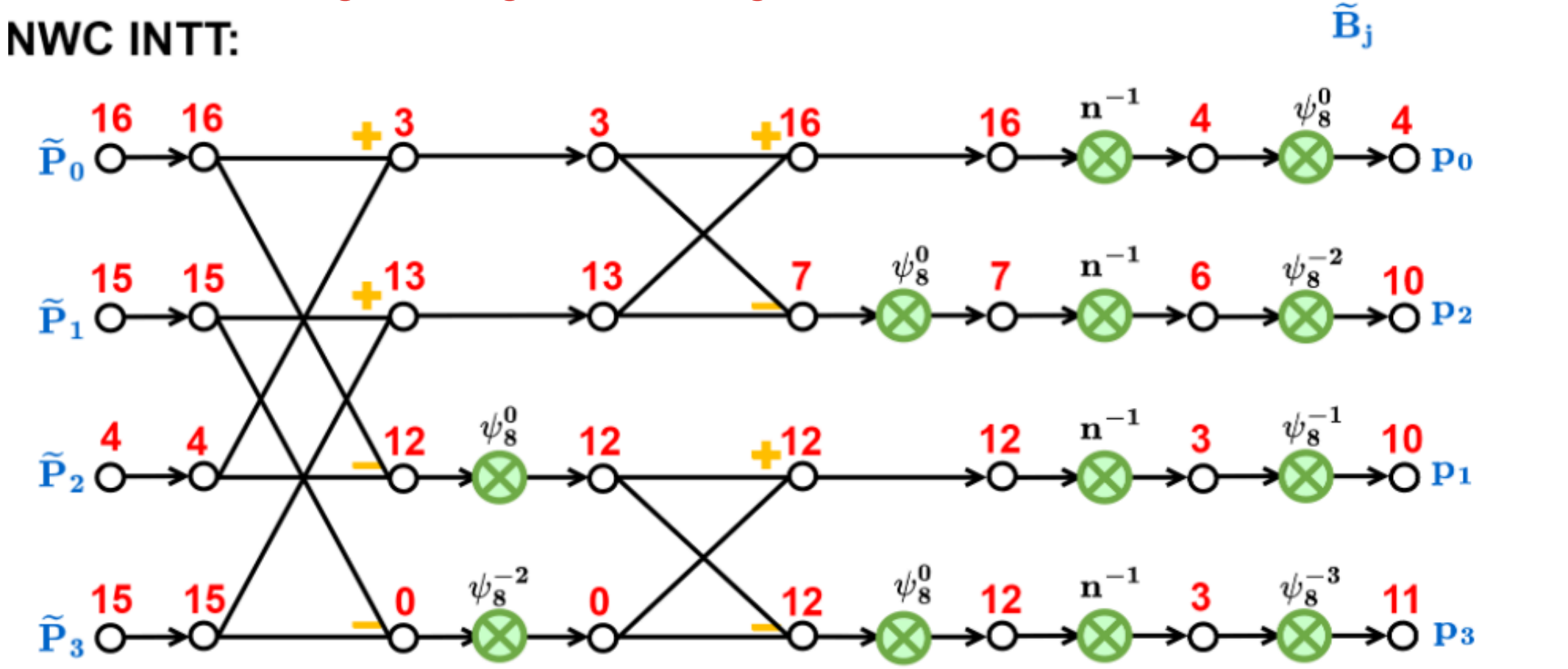
Final Result: $3 + 3x$



Polynomial Multiplication Engine



Cooley Tukey Butterfly for Forward NTT



Gentleman Sende Butterfly for Inverse NTT

Implementation Results

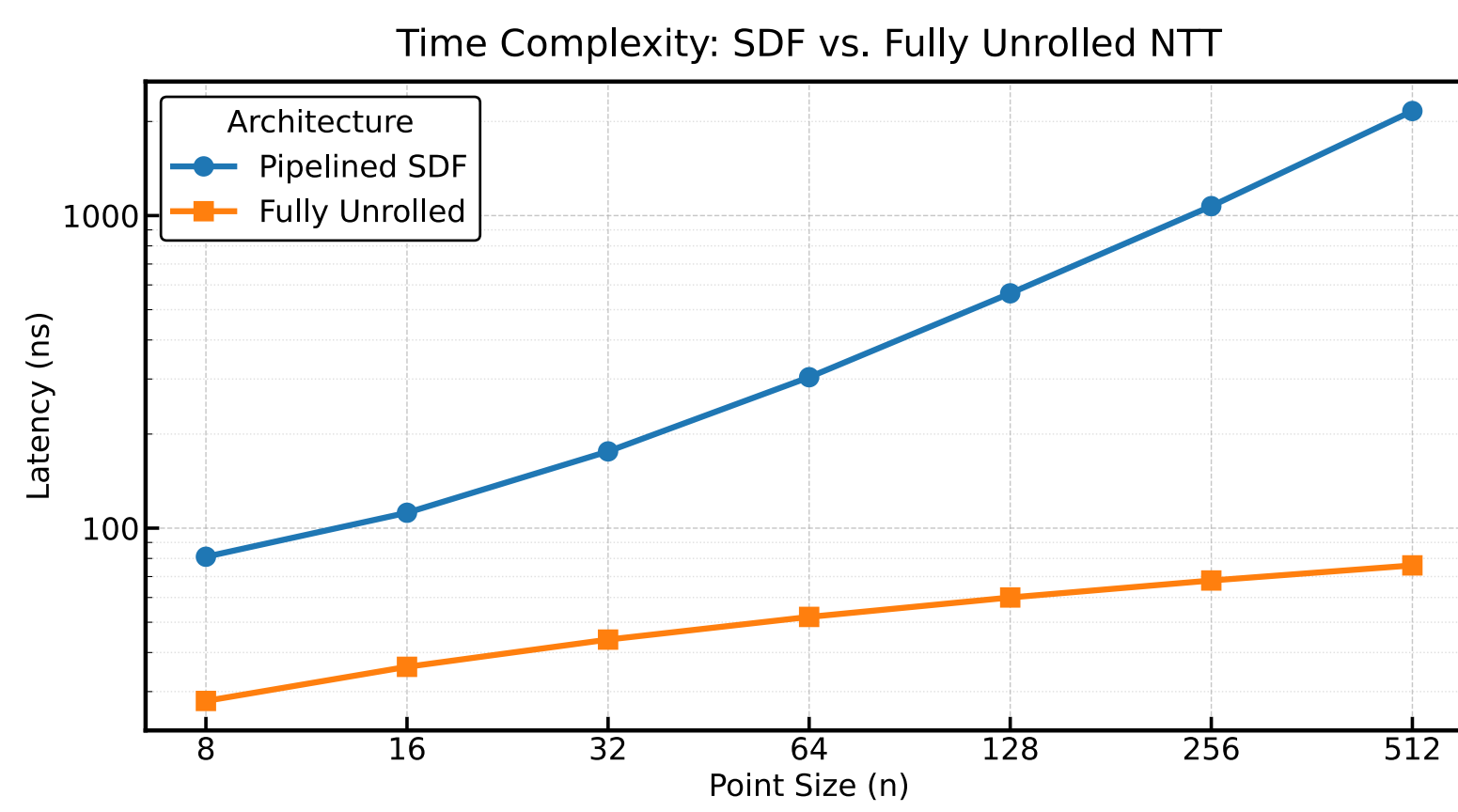


Figure 6: The fully unrolled architecture has a logarithmic growth in latency and SDF has linear.

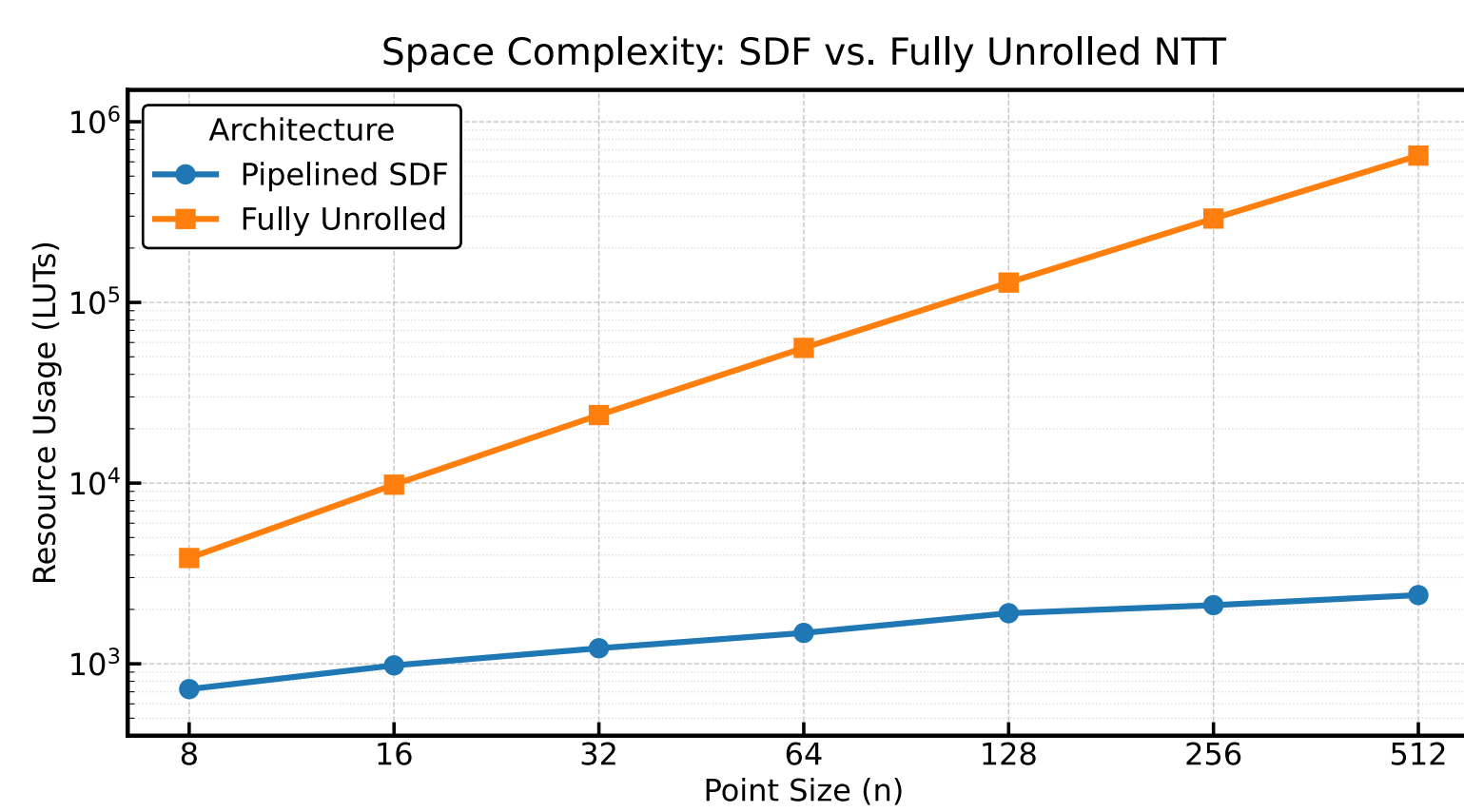


Figure 6: The fully unrolled architecture has a quasilinear growth in space and SDF has logarithmic

Architecture	Space Complexity	Time Complexity
Fully Unrolled	$\mathcal{O}(N \log N)$	$\mathcal{O}(\log N)$
SDF Pipeline	$\mathcal{O}(\log N)$	$\mathcal{O}(N)$

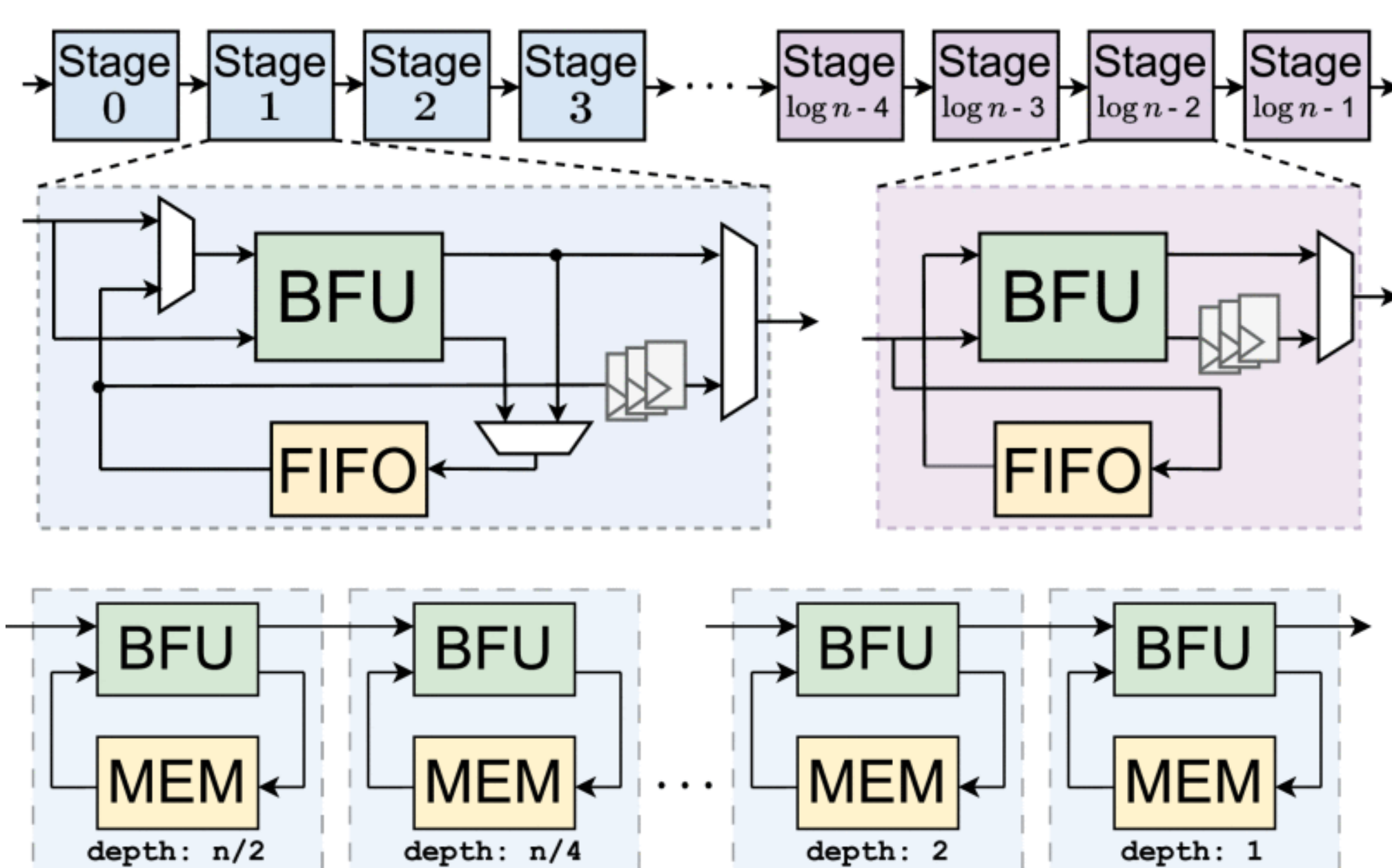
- Hardware Utilization:** The proposed NTT design achieves a significant **area reduction of nearly 20 times compared to Toom-Cook** and up to **75 times compared to TMVP-5**, utilizing only 2400 LUTs at $N = 256$.
- Throughput:** By utilizing a fully pipelined architecture, the design maintains a **constant throughput of 200 M for all values of N**, enabling new inputs to be processed every clock cycle.

$$\mathbb{Z}_5[x] x^2 + 4$$

all numbers mod 5 everything mod $x^2 + 4$

polynomial ring

System Applications and Architecture



- Pipelined SDF:** A streaming Single-Delay Feedback (SDF) - Best
- Performance Goals:** Minimizing Latency (Fully Unrolled), Minimizing Area (SDF)
- Resource Efficiency:** Leveraging BRAMS and DSPs to ensure the design scales efficiently for large polynomial multiplication.

Montgomery modular multiplication

NORMAL DOMAIN

$$C = A \cdot B \pmod{q}$$

The "Guess":

How many times does q fit?

$$\text{Result} = T - (\text{multiple} \cdot q)$$

DIVISION (HIGHER AREA CONSUMPTION)

MONTGOMERY DOMAIN

$$C = \tilde{A} \cdot \tilde{B} \cdot R^{-1} \pmod{q}$$

The "Guess":

What multiple of q clears the bits?

$$\text{Result} = (T + (\text{multiple} \cdot q)) / R$$

ADDITION (LOWER AREA CONSUMPTION)

- Accordingly all the twiddle factors had to be scaled.
- Multiplication with R to exit Montgomery domain

Future Work : Radical Ring LWE

$$A_{s,x} = (a, \langle a, s \rangle + e \pmod{q}) \in \mathbb{R}_q^k \times \mathbb{R}_q \quad [\langle a, s \rangle]_p = \left\lfloor \frac{p}{q} \langle a, s \rangle \right\rfloor \in \mathbb{Z}_p.$$

Learning with errors Learning with rounding

$$\mathbb{Z}_q[x] / ((x^k - 2)^n + 1) \cong (\mathbb{Z}_q[y] / (y^n + 1)[x]) / (x^k - y - 2)$$

INNER RING OUTER RING RELATION

$x^k \mapsto y + 2$ Ring Isomorphism

Key Advantages

- Scalable Security:** Breaks the "power-of-two" constraint of standard Ring-LWE, allowing fine-grained security levels without massive jumps in parameter size.
- Linear Complexity:** Unlike Module-LWE which scales quadratically (k^2) with rank, Radical Ring-LWE maintains linear scaling (k), drastically reducing hardware overhead for high-security targets.

Implementation of Radical Ring LWE:

- Vectorized Polynomial Operations:** Transitioning to a *Module-LWE paradigm* to process math as vectors of smaller rings rather than singular, monolithic polynomials.
- Parameter Agnostic Security:** Offering scalable security targets simply by changing the vector length, thereby increasing time.
- Protocol Deployment:** Utilizing the optimized hardware core to deploy complete PQC scheme.

Conclusions

- Evaluated fully unrolled (minimum latency) and SDF (minimum area) pipelines to establish the absolute operating limits of NTT hardware.
- SDF architecture metrics are comparable with leading edge research designs.
- According to system requirements can now achieve ideal tradeoff between time and space.
- The foundational hardware logic required to efficiently execute these decomposed, nested ring structures is already successfully implemented and verified.

References

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